Lecture 3: Sep 8, 2026

Last time:

- Determining phase angle
- Vertical springs (external constant force)
- Simple pendulum and small angle approximation

Today's class:

- Velocity and acceleration in SHM
- Adding SHMs together (same frequency, different amplitude)
- Review of complex math necessary for phasors

VELOCITY AND ACCELERATION IN SHM

Velocity and acceleration in SHM

- Start with x(t): $x(t) = A\cos(\omega t + \phi_0)$
- Differentiate once to get v(t):

$$v(t) = \frac{dx}{dt} = A[-\omega \sin(\omega t + \phi_0)\,]$$
$$= -A\omega \sin(\omega t + \phi_0)$$

- Maximum speed is: $v_{max} = A\omega$
- Initial conditions: $x(0) = A\cos\phi_0$

$$v(0) = -A\omega \sin\phi_0$$

A and ϕ_0 are constants of integration, so they SHOULD be related to initial conditions

- Differentiate again to get a(t):

$$a(t) = \frac{d^2x}{dt^2} = \frac{dv}{dt}$$
$$= -A\omega[\cos(\omega t + \phi_0)\omega]$$
$$= -A\omega^2 \cos(\omega t + \phi_0)$$

- Maximum acceleration is:

$$a_{max} = A\omega^2$$

- Summary:

$$x(t) = \boldsymbol{A\cos(\omega t + \phi_0)}$$
$$v(t) = -A\omega \sin(\omega t + \phi_0) = \boldsymbol{A\omega \cos(\omega t + \phi_0 + \pi/2)}$$
$$a(t) = -A\omega^2 \cos(\omega t + \phi_0) = \boldsymbol{A\omega^2 \cos(\omega t + \phi_0 + \pi)}$$

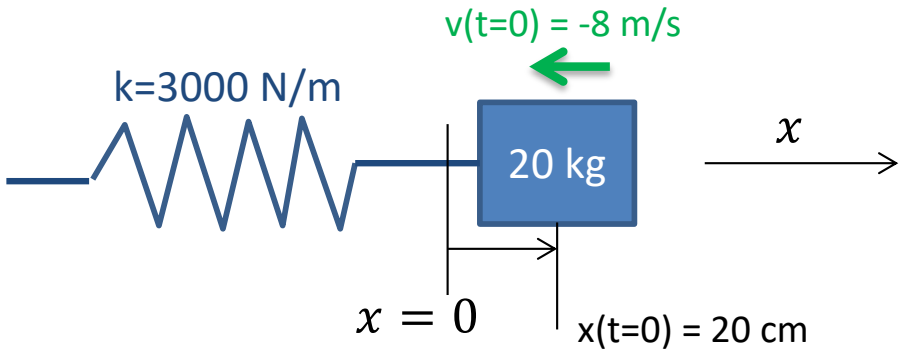
The velocity and acceleration graphs lead the position by $\pi/2$ and π, respectively

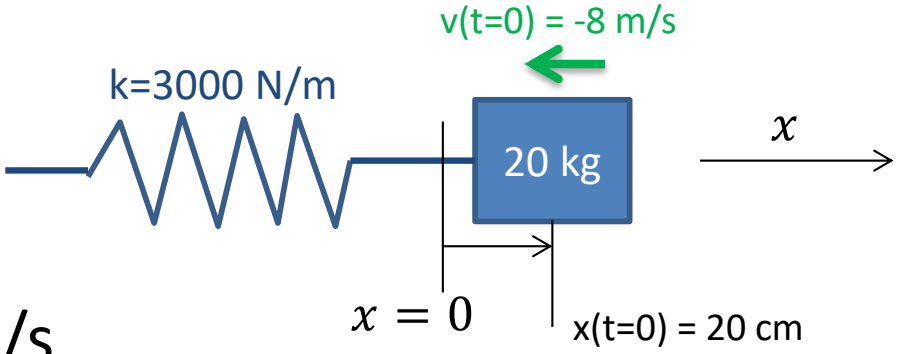
Example

A mass m=20.0 kg is attached to a spring with constant k=3000 N/m.

At t=0, x=+20 cm (measured from equilibrium) and v=-8 m/s.

Find: ω, ϕ_0, A





Find: ω.

Use $\omega^2 = k/m$.

$$\omega = \sqrt{\frac{k}{M}} = \sqrt{\frac{3000\ N/m}{20\ kg}} = 12.247 \text{ rad/s}$$

Find: ϕ_0

Start with basic equations for x,v:

$$\begin{cases} x(t) = A\cos(\omega t + \phi_0) \\ v(t) = -A\omega \sin(\omega t + \phi_0) \end{cases}$$

Use the initial condition at $t = 0$:

$$\begin{cases} x(0) = A\cos(\phi_0) \\ v(0) = -A\omega \sin(\phi_0) \end{cases}$$

Divide the two equations to get:

$$\frac{-A\omega \sin\phi_0}{A\cos\phi_0} = \frac{v(0)}{y(0)} = \frac{-8\ m/s}{20\ cm}$$

$$\omega \tan\phi_0 = 40\ s^{-1}$$

$$\phi_0 = \tan^{-1}\left[\frac{40 s^{-1}}{12.247\ s^{-1}}\right]$$

$$\phi_0 = \tan^{-1}(3.27)$$

But **BE CAREFUL HERE**:

$$\tan^{-1}(3.27) = 1.27 + \boldsymbol{n\pi}$$

How to choose n?

Only one solution fits the signs on the initial conditions:

$$\cos(1.27) = 0.296$$
$$\cos(1.27 \pm \pi) = -0.296$$

The initial position should be positive, so:

$$\phi_0 = 1.27 \text{ rad} = 73°$$

Find: A.

Back-substitute

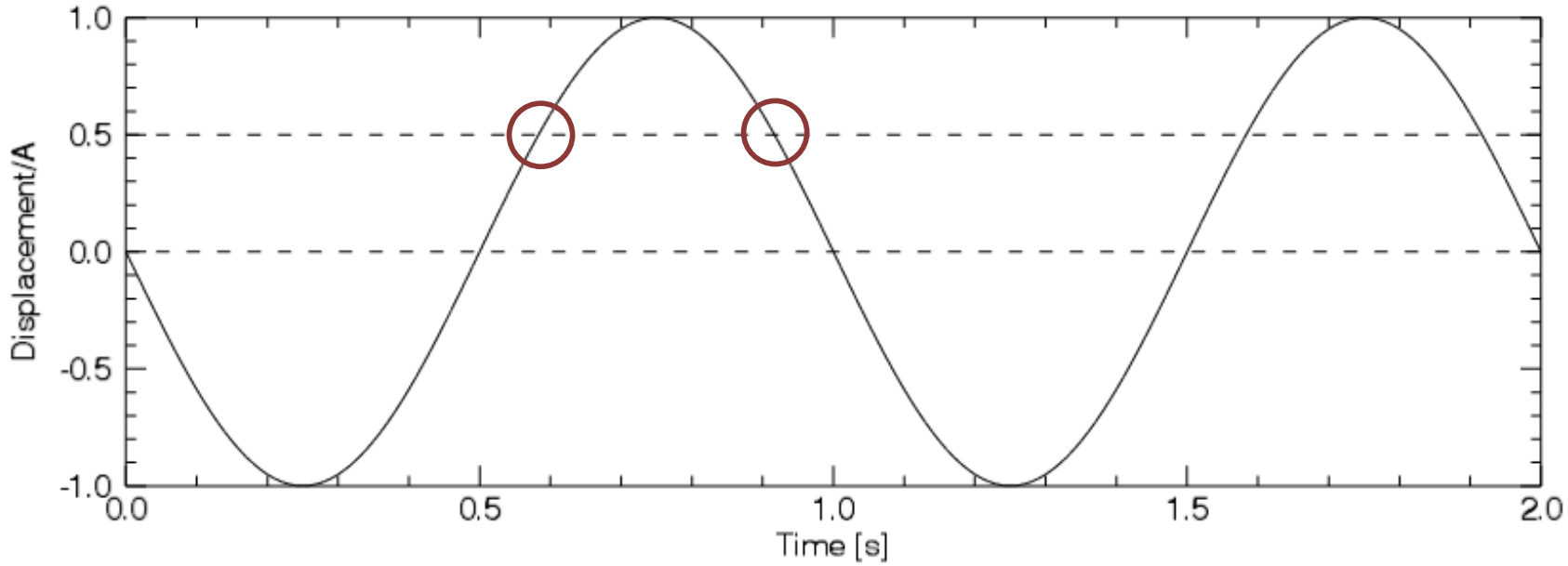
$$x(0) = A\cos(\phi_0)$$

$$A = \frac{x(0)}{\cos(\phi_0)} = \frac{20\ cm}{\cos(1.27)}$$

A = 68 cm

Example

An oscillator with angular frequency $\omega = 2\pi$ starts at position $x = 0$ at time $t = 0$, moving in the $-x$ direction. What are the first <u>two</u> times $t > 0$ that it passes through the position $x = +A/2$?



First, set ϕ_0. Since $x(t=0)=0$, we can represent this using a sine function. To make the initial velocity negative, use

$$x(t) = -A\sin(\omega t)$$

[Or equivalently: $x(t) = A\cos(\omega t + \pi/2)$]

Next, solve for one time (t_a) using your calculator:

$$x(t_a) = -\mathrm{A}\sin(\omega t_a) = A/2$$

$$\omega t_a = \sin^{-1}\left(-\frac{1}{2}\right)$$

$$t_a = -0.0833333\ \mathrm{s}$$

A second time t_b results from the multi-valued sine:

$$\sin^{-1}(x) = \{\theta, \pi - \theta\} + 2\pi n$$

$$\omega t_b = \pi - \sin^{-1}\left(-\frac{1}{2}\right)$$

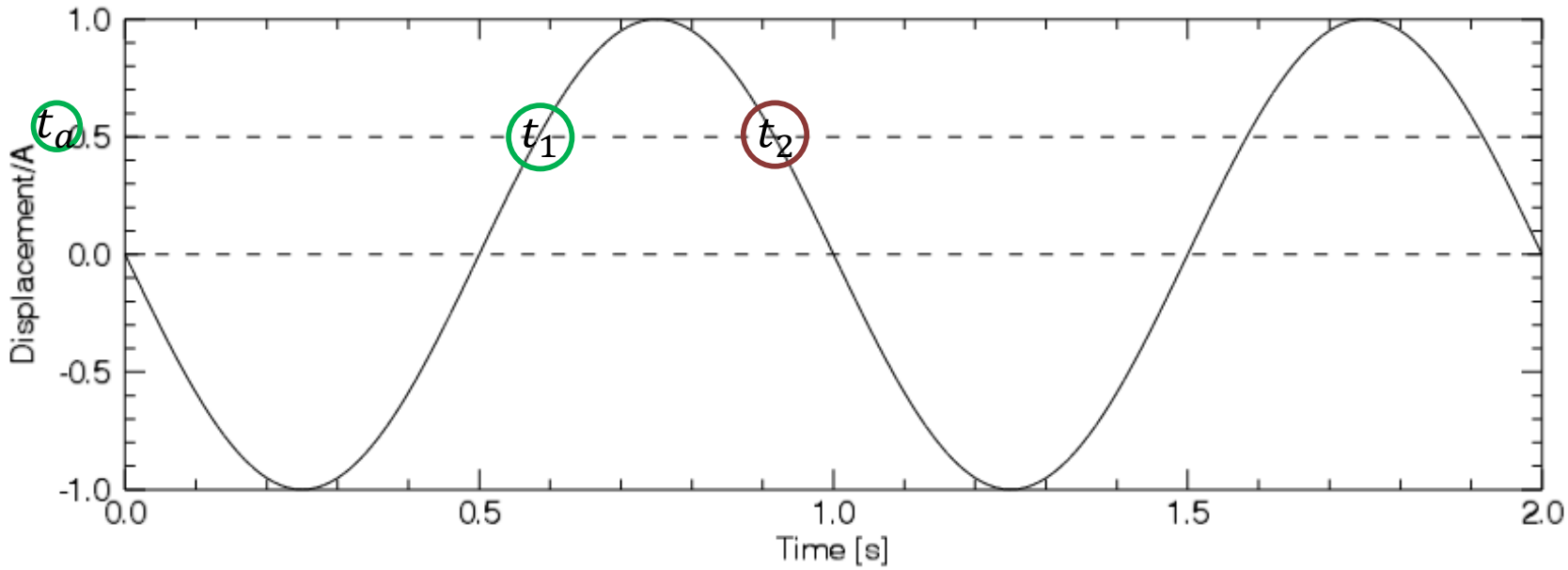
$$t_b = 0.58333$$

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$$t_b = 0.58333$$



Finally, add multiples of the period to find the first two solutions with t>0:

$$t_1 = t_b = 0.583\ \text{s}$$
$$t_2 = t_a + (1\ \text{s}) = 0.917\ \text{s}$$

5-MINUTE BREAK

ADDING SHMS TOGETHER

(OF THE SAME FREQUENCY BUT DIFFERENT AMPLITUDES)

Superposition

- Any oscillation can be broken into a sum (superposition) of simple harmonic oscillations.
- Sum of sinusoids at the same frequency:

$$y = A_1 \cos(\omega t + \phi_1) + A_2 \sin(\omega t + \phi_2)$$

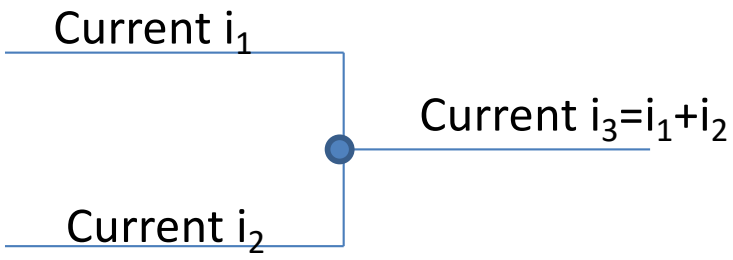
is just another sinusoid:

$$y = A_3 \cos(\omega t + \phi_3)$$

- But A_3 and ϕ_3 may not be easy to find.

Example

Two alternating (AC) currents enter a 3-wire junction. The first is a current of 1.0 Amps at a phase of 30 degrees. The second is 2.0 Amps at a phase of 60 degrees. Find the resulting current out of the junction.



The hard way

$i_3 = A_1 \cos(\omega t + \phi_1) + A_2 \cos(\omega t + \phi_2)$

1. Break apart using the trig identity $\cos(\alpha + \beta) = \cos\alpha cos\beta - \sin\alpha \sin\beta$

$i_3 = A_1 \cos(\omega t) \cos(\phi_1) - A_1 \sin(\omega t) \sin(\phi_1) + A_2 \cos(\omega t) \cos(\phi_2) - A_2 \sin(\omega t) \sin(\phi_2)$

2. Collect terms

$i_3 = [A_1 \cos(\phi_1) + A_2 \cos(\phi_2)]\cos(\omega t) - [A_1 \sin(\phi_1) + A_2 \sin(\phi_2)]\sin(\omega t)$

3. Define A, δ such that:

$$\begin{cases} A\cos\delta = A_1\cos(\phi_1) + A_2\cos(\phi_2) \\ A\sin\delta = A_1\sin(\phi_1) + A_2\sin(\phi_2) \end{cases}$$

4. So that:

$$i_3 = A\cos\delta\cos(\omega t) - A\sin\delta\sin(\omega t) = A\cos(\omega t + \delta)$$

5. Solve the simultaneous equations from step 3 for A, δ

5a. Square and add them, use a few more trig identities:

$$A^2 = A_1^2 + A_2^2 + 2A_1A_2\cos(\phi_2 - \phi_1)$$

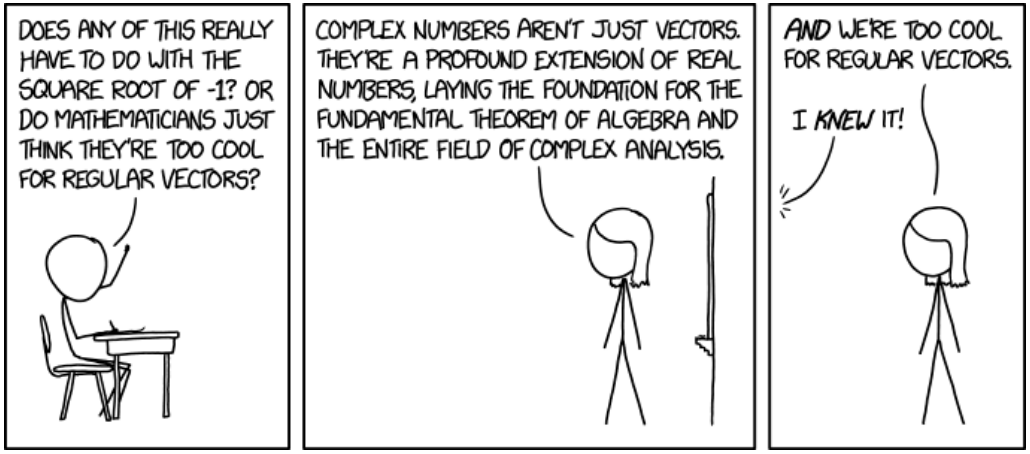
5b. Divide them:

$$\tan\delta = \frac{A_1\sin\phi_1 + A_2\sin\phi_2}{A_1\cos\phi_1 + A_2\cos\phi_2}$$

6. Collect:

$$i_3 = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos(\phi_2 - \phi_1)}\ \cos\left[\omega t + \tan^{-1}\left(\frac{A_1\sin\phi_1 + A_2\sin\phi_2}{A_1\cos\phi_1 + A_2\cos\phi_2}\right)\right]$$

7. Plug in values.



https://xkcd.com/2028/

COMPLEX MATH REVIEW

Complex math

We'll use j instead of i for the imaginary unit $\sqrt{-1}$. Blame the electrical engineers.

If this isn't feeling familiar/comfortable, see the review on D2L (under Content → Course information and resources → Extra notes).

- Addition:
- Product:

Complex math

- Complex number

$$z = x + jy$$

$$j = \sqrt{-1}$$

x: real part

y: imaginary part

- Addition: $z_1 \pm z_2 = (x_1 \pm x_2) + j(y_1 \pm y_2)$
- Product: $z_1 z_2 = (x_1 + jy_1)(x_2 + jy_2)$

$$= (x_1 x_2 - y_1 y_2) + j(x_1 y_2 + y_1 x_2)$$

We'll use j instead of i for the imaginary unit $\sqrt{-1}$. Blame the electrical engineers.

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- Complex conjugate:
- Magnitude (modulus, absolute value):
- Complex exponential (Euler's relation):

- Complex conjugate:

$$z^* = x - jy$$

- Magnitude (modulus, absolute value):

$$|z| = \sqrt{x^2 + y^2} = \sqrt{zz^*}$$

- Complex exponential (Euler's relation):

$$e^{j\theta} = cos\theta + j\sin\theta$$